

# Supplementary Material for Backbone-Modular Neuro-Symbolic Post-Correction for OCT Denoising

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This document holds the material that does not fit inside the page limit of the main paper. It adds nothing to the claims. It gives the closed form of every clinical property, the proof of the second theorem, the training algorithm, the full list of constants, the visual comparison for all four backbones, and a note on a defect found and repaired during revision.

## S1. Closed form of the six clinical properties

Table S1 gives the score and the failure map of every property, and the expanded definitions of every subterm follow. Fig. 2 of the main paper shows all six maps on one image. All quantities are computed from the noisy input and the backbone output alone. No clean reference is used at any point.

The release computes exactly these quantities. The relevant class is named `EnhancedGTFreePredicates` and the speckle property is computed by `PhysicsAccurateSpecklePredicate`.

### A. Expanded subterms of Table 2 of the main paper

Subterms.  $P_1$ .  $c_{\text{cont}} = \langle \text{erode}(\text{dilate}(b_e)), b_e \rangle / (\|b_e\|_1 + \varepsilon)$  carries no information and we say so rather than leave a reader to derive it. Erosion after dilation by the same  $3 \times 3$  element is a morphological closing, which contains the set it closes, so the inner product equals  $\|b_e\|_1$  and the ratio is one wherever the closing is defined. Measured on  $256 \times 256$  fields it reads 0.952, and every pixel of the shortfall lies on the two pixel border, where the zero padding erodes the dilated set. Restricted to the interior it is exactly 1. The subterm therefore contributes a constant near 0.4 to  $P_1$  and the score varies only through  $c_{\text{efr}}$  and  $c_{\text{ang}}$ . We are careful about what follows from that. The failure map of  $P_1$  does not read this subterm, so it does not enter the allocation through the per pixel path. It does enter  $s_1$ , which reaches the soft failure of Equation (3) and the acceptance energy of Equation (11), so it is not inert in the forward computation and we do not claim the output is independent of it. What we claim is narrower, that it is non discriminative, a near constant offset rather than a measurement of continuity. We keep it exactly as it was run, since replacing it would shift  $s_1$  and could change a rule truth value or an acceptance

decision, and every number in this paper was produced with it in place. The other two subterms are  $c_{\text{efr}} = \exp(-5|r_e - 0.2|)$  with  $r_e = \|\mathbf{1}[e > 1.5\bar{e}]\|_1 / (\|\mathbf{1}[e < 0.5\bar{e}]\|_1 + \varepsilon)$ , and  $c_{\text{ang}} = \exp(-2 \text{mean}(\sigma_5(\theta)))$  with  $\theta$  the Sobel edge angle of  $\mathbf{b}$ .  $P_2$ .  $s_{\text{range}} = \text{sigm}(10(\Delta_r - 0.3))$  with  $\Delta_r = \max(\mathbf{b}) - \min(\mathbf{b})$ ,  $s_{\text{cnr}} = \text{sigm}(15(\bar{\sigma} - 0.05))$ ,  $s_{\text{var}} = \max(0.2, \text{sigm}(-20(v_{\sigma} - 0.02) + 2))$  with  $v_{\sigma} = \text{Var}(\sigma_9(\mathbf{b}))$ , and  $s_{\text{cov}} = \text{sigm}(5(\gamma - 0.4))$  with  $\gamma = \text{mean}(\mathbf{1}[\sigma_9(\mathbf{b}) > 0.3\bar{\sigma}])$ .  $P_3$ .  $F = \mathbf{1}[e < 0.3\bar{e}]$ ,  $q_{\text{var}} = \text{clip}(1 - \langle F, \sigma_5(\mathbf{b}) / (\sigma_5(\mathbf{y}) + \varepsilon) \rangle / (\|F\|_1 + \varepsilon), 0, 1)$ , the clip being necessary because the variance ratio may exceed one and leave the subterm negative, and  $q_{\text{res}} = \exp(-5 \text{mean}|\mathbf{y} - \mathbf{b}|)$ .  $P_4$ .  $s_{\text{self}}$  is the mean Pearson correlation between  $\mathbf{b}$  and its shifts by  $(0, 2)$ ,  $(2, 0)$  and  $(2, 2)$  pixels, each clipped to  $[-1, 1]$  and the mean then clipped to  $[0, 1]$ , since a correlation may be negative,  $s_{\text{reg}} = \exp(-12 \text{mean}(e(e)))$ , and  $s_{\text{vert}} = \text{clip}(\max|\nabla_z v(\mathbf{b})| / (8 \text{mean}|\nabla_z v(\mathbf{b})|), 0, 1)$ .  $P_5$ .  $z = \log(\mathbf{y} + \varepsilon) - \log(\mathbf{b} + \varepsilon)$ ,  $cv = \sigma_{15}(|z|) / (\mu_{15}(|z|) + \varepsilon)$  with  $\mu_{15}$  the local mean, and  $cv^*(d) = 1/\sqrt{k(d)}$  with  $k(d)$  taken piecewise in relative axial depth as  $k = 2.5$  above 0.20, 4.0 to 0.50, 10.0 to 0.75 and 1.5 below, giving targets of 0.63, 0.50, 0.32 and 0.82. We are careful about what this target is.  $1/\sqrt{k}$  is the coefficient of variation of a Gamma variable of shape  $k$ , which is the standard model of fully developed speckle intensity, whereas  $cv$  above is measured on  $|z|$ , the absolute log domain residual. A coefficient of variation does not survive that nonlinear transform, so  $cv^*$  is a speckle motivated target with the right depth ordering and not a quantity derived from the Gamma model. It sets where the property is satisfied, and the depth dependence is the part of it that carries the physics.  $P_6$ .  $s_{\text{clr}} = \text{mean}(\text{clip}(\max|\nabla_z v(\mathbf{b})| / (10 \text{mean}|\nabla_z v(\mathbf{b})|), 0, 1))$ ,  $s_{\text{dep}} = \text{clip}(20 \text{Var}(v(\mathbf{b})), 0, 1)$ , and  $s_{\text{smo}} = 1 - \text{clip}(10 \text{mean}(\text{Var}_z(\text{mean}_w e)), 0, 1)$  with  $\text{mean}_w$  the mean across width and  $\text{Var}_z$  the variance along depth.

## S2. Proofs

### A. Proof of the allocation stability result

This is the proof of Theorem 1 of the main paper (Allocation stability), repeated here in full. Write  $s_k = -1$  for  $k \in \{1, 5\}$  and  $s_k = +1$  otherwise, so that the allocation equation of the main paper reads

**TABLE S1.** Closed form of the six clinical properties. For each predicate, the table lists its clinical role, image-level score, and pixel-level failure map. larger  $\mathbf{m}_i$  means stronger evidence that the corresponding property is locally unsatisfied. The expanded subterm definitions follow immediately below. Here  $\bar{e} = \text{mean}(e(\mathbf{b}))$ ,  $b_e = \mathbf{1}[e(\mathbf{b}) > \bar{e}]$ ,  $v(z) = \text{mean}_w z$  is a depth profile,  $\rho_h$  is 1-pixel horizontal autocorrelation, and  $\text{Var}_k$  is local variance over a  $k \times k$  window.

| $P_i$ | Clinical role           | Aggregate score $s_i$                                                                     | Failure map $\mathbf{m}_i$                                                                                      | $\tau_i$ |
|-------|-------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------|
| $P_1$ | Edge continuity         | $0.4 c_{\text{cont}} + 0.3 c_{\text{efr}} + 0.3 c_{\text{ang}}$                           | $(1 - \bar{e})(1 - b_e)$ , $\bar{e} = e(\mathbf{b})/(e_{\text{max}} + \varepsilon)$                             | 0.6      |
| $P_2$ | Local contrast recovery | $0.20 s_{\text{range}} + 0.35 s_{\text{cnr}} + 0.20 s_{\text{var}} + 0.25 s_{\text{cov}}$ | $[\text{ReLU}(\bar{\sigma} - \sigma_9)/(\bar{\sigma} + \varepsilon) + 0.3 \mathbf{1}[\sigma_9 < 0.01]]_{[0,1]}$ | 0.5      |
| $P_3$ | Flat-region smoothness  | $0.6 q_{\text{var}} + 0.4 q_{\text{res}}$                                                 | $F \odot \sigma_5(\mathbf{b})/(\max \sigma_5(\mathbf{b}) + \varepsilon)$                                        | 0.75     |
| $P_4$ | Structural coherence    | $0.4 s_{\text{self}} + 0.3 s_{\text{reg}} + 0.3 s_{\text{vert}}$                          | $e(e(\mathbf{b})) / (\max e(\mathbf{b}) + \varepsilon)$                                                         | 0.5      |
| $P_5$ | Speckle conformity      | $0.7 \text{mean}(\exp(- cv - cv^* /0.15)) - 0.3 \exp(-5 \rho_h(z) )$                      | $+ 1 - \exp(- cv - cv^* /0.15)$                                                                                 | 0.3      |
| $P_6$ | Layer visibility        | $0.4 s_{\text{clr}} + 0.3 s_{\text{dep}} + 0.3 s_{\text{sno}}$                            | $\text{Var}_7(e(\mathbf{b})) / (\max \text{Var}_7(e(\mathbf{b})) + \varepsilon)$                                | 0.7      |

$a(\mathbf{r}) = b + \sum_k s_k C_k w_k r_k$  with  $b$  independent of  $\mathbf{r}$ , where  $\mathbf{r} = (r_1, \dots, r_5) \in [0, 1]^5$  is the vector of rule truth values at one pixel,  $C_k$  are the fixed budgets, and  $w_k = \sigma(\theta_k)$  are the learned rule weights.

Proof:

Subtracting two such expressions gives

$$a(\mathbf{r}) - a(\mathbf{r}') = \sum_k s_k C_k w_k (r_k - r'_k).$$

Taking absolute values and applying the triangle inequality,

$$\begin{aligned} |a(\mathbf{r}) - a(\mathbf{r}')| &\leq \sum_k C_k w_k |r_k - r'_k| \\ &\leq \left( \sum_k C_k w_k \right) \|\mathbf{r} - \mathbf{r}'\|_\infty, \end{aligned}$$

which is the stability bound of Definition 1 of the main paper with  $L = \sum_k C_k w_k$  as claimed. The bound is attained by taking  $r_k - r'_k = s_k t$  for any  $t$  small enough to keep both vectors in the unit cube, since then every term of the sum has the same sign. Because each  $C_k$  is positive and each  $w_k = \sigma(\theta_k)$  lies strictly in  $(0, 1)$ , the sum is strictly less than  $\sum_k C_k = 0.75$ . ■

#### B. Proof of the contrast margin result

This is the proof of Theorem 2 of the main paper, repeated here in full. Write  $\mu_{\mathcal{T}}$  and  $\mu_{\mathcal{B}}$  for the mean over tissue and over background, and  $\sigma_{\mathcal{B}}$  for the standard deviation over background. Let  $\mathbf{b}$  be the backbone output,  $\mathbf{q}$  the candidate, and  $\hat{\mathbf{x}} = (1 - w)\mathbf{b} + w\mathbf{q}$  the output with  $w \in (0, 1]$ .

Theorem S1 (Contrast margin). Let  $\eta_{\mathcal{T}} = \mu_{\mathcal{T}}(\mathbf{q}) - \mu_{\mathcal{T}}(\mathbf{b}) \geq 0$  and let  $\eta_{\mathcal{B}} \geq \mu_{\mathcal{B}}(\mathbf{q}) - \mu_{\mathcal{B}}(\mathbf{b})$ . If  $\sigma_{\mathcal{B}}(\mathbf{q}) \leq \sigma_{\mathcal{B}}(\mathbf{b})$ , that  $\sigma_{\mathcal{B}}(\mathbf{b}) > 0$  and  $\sigma_{\mathcal{B}}(\hat{\mathbf{x}}) > 0$  so that both ratios are defined, that the tissue and background sets are the same for all three images, and that the bounded numerator is

non negative, then

$$\text{CNR}(\hat{\mathbf{x}}) \geq \frac{(\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta_{\mathcal{B}})}{\sigma_{\mathcal{B}}(\mathbf{b})}. \quad (\text{S1})$$

Proof:

The mean is linear, so over any fixed set of pixels the mean of a convex combination is the same convex combination of the means. Applying this over the tissue set and over the background set,

$$\begin{aligned} \mu_{\mathcal{T}}(\hat{\mathbf{x}}) &= (1 - w)\mu_{\mathcal{T}}(\mathbf{b}) + w\mu_{\mathcal{T}}(\mathbf{q}) = \mu_{\mathcal{T}}(\mathbf{b}) + w\eta_{\mathcal{T}}, \\ \mu_{\mathcal{B}}(\hat{\mathbf{x}}) &= \mu_{\mathcal{B}}(\mathbf{b}) + w\eta'_{\mathcal{B}}, \end{aligned}$$

where  $\eta'_{\mathcal{B}} = \mu_{\mathcal{B}}(\mathbf{q}) - \mu_{\mathcal{B}}(\mathbf{b}) \leq \eta_{\mathcal{B}}$  by hypothesis. Subtracting,

$$\begin{aligned} \mu_{\mathcal{T}}(\hat{\mathbf{x}}) - \mu_{\mathcal{B}}(\hat{\mathbf{x}}) &= (\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta'_{\mathcal{B}}) \\ &\geq (\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta_{\mathcal{B}}), \end{aligned}$$

which lower bounds the numerator.

For the denominator, every background pixel of  $\hat{\mathbf{x}}$  is the same convex combination of the corresponding pixels of  $\mathbf{b}$  and  $\mathbf{q}$ . Treating the background pixel values as a random variable and applying Minkowski's inequality to the centered variables,

$$\begin{aligned} \sigma_{\mathcal{B}}(\hat{\mathbf{x}}) &\leq (1 - w)\sigma_{\mathcal{B}}(\mathbf{b}) + w\sigma_{\mathcal{B}}(\mathbf{q}) \\ &\leq (1 - w)\sigma_{\mathcal{B}}(\mathbf{b}) + w\sigma_{\mathcal{B}}(\mathbf{b}) = \sigma_{\mathcal{B}}(\mathbf{b}), \end{aligned}$$

using the hypothesis  $\sigma_{\mathcal{B}}(\mathbf{q}) \leq \sigma_{\mathcal{B}}(\mathbf{b})$  in the second step. Dividing the lower bound on the numerator by the upper bound on the denominator gives the claim. Since  $w > 0$  in every branch of the blending rule, the bound is at least the baseline contrast whenever  $\eta_{\mathcal{T}} > \eta_{\mathcal{B}}$ . ■

The results section of the main paper reports how often each of the three hypotheses holds on the test set, so the reader can see when the guarantee is active rather than assuming it always is.

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**Algorithm 1** Training the wrapper.

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Require: frozen backbone  $f_\theta$ , paired training set, number of epochs  $E$

- 1: initialize the wrapper parameters, set each property threshold to its literature value
- 2: for  $e = 1$  to  $E$  do
- 3:   for each batch of crops do
- 4:     compute the backbone output, with no gradient to  $\theta$
- 5:     compute the cooperation map from the backbone features
- 6:     compute the six property scores and failure maps
- 7:     compute the gain, edge and strength proposals
- 8:     compute the five rule truth values
- 9:     compute the allocation
- 10:    compose the candidate  $\mathbf{q}$ , then evaluate the objective on it
- 11:     $\triangleright$  the acceptance rule of the main paper does not run here
- 12:    update the wrapper parameters with AdamW
- 13:   end for
- 14:   if  $e$  reaches the midpoint then activate the fidelity term
- 15:   end if
- 16: end for

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**S3. Training procedure and the full list of constants****A. Algorithm**

Algorithm 1 states the procedure. The backbone is frozen throughout and receives no gradient. The objective is evaluated on the candidate, because the acceptance rule is skipped while the model is in training mode, so Equation (1) of the main paper is optimized at  $\mathbf{q}$  and not at  $\hat{\mathbf{x}}$ . The first stage activates only the clinical terms, so the correctors learn a useful direction. The second stage activates the fidelity term, so they learn how far to go.

**B. Schedule and optimizer settings**

We use AdamW with decoupled weight decay of  $10^{-4}$ , random crops of  $96 \times 96$  pixels, and a batch of sixteen crops, reduced to eight for DnCNN and SwinIR, the two with the largest memory footprint. Three learning rates are used, a lower one for the corrector networks and higher ones for the strength predictor and the rule layer, because the last two have far fewer parameters and therefore need larger steps to move at all. The switch from the clinical only stage to the stage that also includes fidelity happens at the midpoint of the schedule, matching the epoch check in Algorithm 1.

**C. The nine quantities of the clinical term**

Equation (15) of the main paper names nine quantities and their weights. Their definitions are here, so the term can be rebuilt. Write  $\sigma_7$  for the local standard deviation over a  $7 \times 7$  window,  $\nabla_z$  and  $\nabla_x$  for the vertical and horizontal Sobel responses,  $\nabla^2$  for the Laplacian,  $e = \sqrt{\nabla_x^2 + \nabla_z^2 + \varepsilon}$  for the edge magnitude, and  $\langle \cdot \rangle$  for the mean over every pixel of the batch. All nine are scalars per batch. The first four are ratios against the clean reference  $\mathbf{x}$ . CNR, ENL and SNR are computed from fixed bands of the frame and use no reference, BS is a ratio of maxima against the reference, and the last is a squared error ratio against it. These are the training time quantities and are not the anatomically masked measures used to score results in the main paper.

- Local contrast,  $\langle \sigma_7(\hat{\mathbf{x}}) \rangle / \langle \sigma_7(\mathbf{x}) \rangle$ .
- Boundary gradient,  $\langle |\nabla_z \hat{\mathbf{x}}| \rangle / \langle |\nabla_z \mathbf{x}| \rangle$ .
- Texture,  $\langle |\nabla^2 \hat{\mathbf{x}}| \rangle / \langle |\nabla^2 \mathbf{x}| \rangle$ .
- EPI, the Pearson correlation between  $e(\hat{\mathbf{x}})$  and  $e(\mathbf{x})$ , taken per image over all pixels and then averaged over the batch.
- CNR,  $(\mu_U - \mu_R) / \sigma_R$  on the corrected image, with  $U$  the upper half and  $R$  the lower quarter of the frame.
- BS,  $\max |\nabla_z \hat{\mathbf{x}}| / \max |\nabla_z \mathbf{x}|$ , clipped to  $[0, 10]$ , then divided by the same ratio for the backbone.
- ENL,  $(\mu_R / \sigma_R)^2$ , and SNR,  $\mu_U / \sigma_R$ , each as the exponential of the mean log ratio to the backbone value over the batch.
- The squared error ratio,  $\|\mathbf{b} - \mathbf{x}\|_2^2 / \|\hat{\mathbf{x}} - \mathbf{x}\|_2^2$ .

Every denominator is clamped away from zero at  $10^{-6}$ . The target  $t_m$  of each is the same quantity on the backbone output, held with no gradient and scaled by the factor  $\kappa_m$  the main paper lists, except ENL, SNR and the squared error ratio, whose targets are the fixed values 1.10, 1.15 and 0.87 rather than backbone multiples.

**D. The selection rule in full**

The main paper says a setting is admissible only at its worst seed and that the winner has the largest selection score on its weakest clinical measure. Both are stated here exactly, since they decide the reported operating point.

For a setting with  $n$  seeds, let  $m_s = \min_i \Delta_i^{(s)}$  be the weakest of the six clinical changes at seed  $s$ . The score it is ranked by is

$$S = \frac{1}{n} \sum_s m_s - \frac{1}{\sqrt{n}} \text{sd}(m_1, \dots, m_n), \quad (\text{S2})$$

the mean of the worst measure less one standard error of that mean, and the mean itself when  $n = 1$ . We call it a selection score rather than a confidence bound, since no coverage level is claimed and three seeds would not support one. A setting is admissible at a level  $(c, f)$

when its fidelity loss at its worst seed is at most  $c$  decibels and its weakest clinical change at its worst seed exceeds  $f$  percent. The four levels are applied in order, strict (0.8, 0.5), standard (1.0, 0.2), positive (1.0, 0.0) and unbounded ( $\infty$ , 0.0). The first level that admits any setting decides the choice, and the level used is recorded, so a backbone that needed a looser condition is visible rather than hidden. Among the admissible settings the winner is the largest score, with the mean fidelity and then the mean over the six measures breaking ties. If no level admits any setting the script reports the setting failing on the fewest measures and labels the outcome as no selection.

That last branch is the one two backbones took. NAFNet and SwinIR were admitted at the strict level, with scores of +1.15 and +1.77. No setting in the grid cleared any of the four levels for DnCNN or KBNNet, whose best scores were  $-1.95$  and  $-0.05$ , so for those two the reported setting is the least bad of the grid and not a setting that met a stated bound. They are also the two backbones with a clinical measure that falls on test, which is consistent with a grid that had nothing admissible to offer. Seeds 0, 1 and 2 are used throughout, and the seed 0 checkpoint of the selected setting is the one every figure draws.

#### E. Constants that carry a design decision

Table S2 lists the constants that carry a design decision, where each acts, and how it was chosen. A constant marked measured was set from a property of the data. A constant marked selected was chosen on the validation split and never on the test split. A constant marked fixed is a structural choice that was not tuned.

The networks use the default initialization of their layers. Two exceptions are deliberate. The bias of the brightness head starts at a small positive value so that the multiplicative branch begins from a mild brightening rather than from zero, which avoids a flat region of the loss at the start of training. The threshold of each property is declared as a trainable parameter and set to the value reported in the clinical literature for that property. It receives no gradient under the objective of the main paper, and in every selected checkpoint all six sit at their literature values to four decimals, so they are fixed in effect and the main paper reports them as fixed.

#### F. Structural constants

Table S3 lists the remaining constants, which are structural and were not tuned.

#### G. Selected gate rule and dead zone per backbone

Table S4 lists, for each backbone, the gate rule chosen by the search described in the training section of the main paper, the radius at which a tissue-bounded gate applies, and the selected fidelity dead zone.

#### H. What the paired data is used for

Training uses paired data. The clean reference enters the fidelity term, the clinical term and the supervision of the edge branch. Testing uses no clean image anywhere in the correction path. The six properties are computed from the noisy input and the backbone output only, which is what allows them to run on a scanner the method has never seen. The clean reference is used to score results, as in any denoising study, and never to produce them.

#### S4. A defect found during revision, and its repair

We record this because the released code contains a switch that restores the earlier behavior, and a reader who finds that switch deserves to know what it is for.

##### A. What was wrong

In the version of this work that was first submitted, the allocation was formed by adding the rule contributions to a base level and then clipping the result to the unit interval. The positive contributions could reach 0.884 and 0.762 on top of a base of 0.818, so the value arriving at the clip could be as large as 2.46. The clip was therefore active at every pixel of every image and the allocation was the constant one.

Two consequences follow. The rule layer had no influence on any output, since multiplying by one changes nothing. And the derivative of a clip at its bound is zero, so no gradient reached the rule parameters and they stayed at their initial values for the whole of training. We confirmed both directly, by reading the allocation map out of the trained model and by comparing every rule parameter in the saved weights against its initializer after 2082 optimizer steps.

##### B. The repair

The allocation is now a bounded weighted vote. Writing  $r_k \in [0, 1]$  for the truth value of rule  $k$  and  $w_k \in (0, 1)$  for its learned weight,

$$\begin{aligned} a = & B_{\min} + (B_{\max} - B_{\min}) \sigma(\beta) \\ & - C_1 w_1 r_1 + C_2 w_2 r_2 + C_3 w_3 r_3 \\ & + C_4 w_4 r_4 - C_5 w_5 r_5, \end{aligned} \quad (\text{S3})$$

with  $B_{\min} = 0.25$ ,  $B_{\max} = 0.50$  and budgets  $C_1 = 0.15$ ,  $C_2 = 0.25$ ,  $C_3 = 0.15$ ,  $C_4 = 0.10$  and  $C_5 = 0.10$ . The budgets satisfy

$$B_{\max} + C_2 + C_3 + C_4 = 1, \quad B_{\min} - C_1 - C_5 = 0, \quad (\text{S4})$$

so the value lies in the unit interval before any clipping, for every input and every parameter setting. Clipping is therefore never active and saturation is impossible rather than unlikely, which restores the gradient to the rule layer. It restores it to four of the five weighted rules and to the base. The fifth is a separate matter and not a consequence of the clamp. Its truth value is computed only where more than one corrector proposes at a pixel,

TABLE S2. Every constant in the method, where it acts and how it was chosen. Fixed means a structural choice that was not tuned. Set in development means it was chosen once on the validation split while the method was being built and never revisited for this revision. Selected in the revision means it entered the grid search of the revision, on the validation split and never on the test split. Learned means it is a parameter of the model fitted by the objective. No constant was ever chosen on the test split.

| Constant                         | Value                                                  | Acts on                           | Origin                   |
|----------------------------------|--------------------------------------------------------|-----------------------------------|--------------------------|
| $\alpha_{\max}$                  | 0.15                                                   | scale of the raw gain             | set in development       |
| gain bound                       | $\pm 0.20$                                             | Eq. (7) of the main paper         | set in development       |
| gate percentile                  | 0.15                                                   | Eq. (8) of the main paper         | set in development       |
| $\kappa$                         | 10                                                     | Eq. (3) of the main paper         | fixed                    |
| $B_{\min}, B_{\max}$             | 0.25, 0.50                                             | base of Eq. (4) of the main paper | fixed by budget          |
| $C_1 \dots C_5$                  | 0.15, 0.25, 0.15, 0.10, 0.10                           | rule votes                        | fixed by budget          |
| $\tau_1 \dots \tau_6$            | 0.6, 0.5, 0.75, 0.5, 0.3, 0.7                          | Eq. (3) of the main paper         | fixed                    |
| $\beta, \theta_1 \dots \theta_5$ | learned                                                | Eq. (4) of the main paper         | learned                  |
| energy tolerance                 | 0.10                                                   | constraint one                    | set in development       |
| Pareto tolerance                 | 0.10                                                   | constraint two                    | set in development       |
| change bound                     | 0.2                                                    | constraint three                  | set in development       |
| background blend                 | 0.5                                                    | background smoothing              | set in development       |
| learning rates                   | $2 \times 10^{-4}, 5 \times 10^{-4}, 3 \times 10^{-4}$ | training                          | set in development       |
| fidelity dead zone $d$           | 0.6 or 0.9 dB                                          | Eq. (14) of the main paper        | selected in the revision |
| gate rule                        | percentile or tissue bounded                           | Eq. (8) of the main paper         | selected in the revision |

TABLE S3. Structural constants. None of these was tuned.

| Constant           | Value                         | Role                                     |
|--------------------|-------------------------------|------------------------------------------|
| gate steepness     | 20                            | sharpness of the background gate         |
| failure sharpness  | 10                            | sharpness of score to failure conversion |
| smoothing window   | $3 \times 3$                  | smoothing of the brightening term        |
| erosion window     | $5 \times 5$                  | keeps background smoothing off tissue    |
| crop size          | $96 \times 96$                | training crops                           |
| batch size         | 16, or 8 for DnCNN and SwinIR | training batch                           |
| edge bound         | $\pm 0.08$                    | nominal bound of the additive branch     |
| quarter resolution | $H/4 \times W/4$              | scale of the gain network                |

TABLE S4. The gate rule and fidelity dead zone selected for each backbone. The radius column applies only to the tissue-bounded gate.

| Backbone | Gate rule      | Radius         | Dead zone |
|----------|----------------|----------------|-----------|
| NAFNet   | tissue bounded | 41             | 0.6       |
| DnCNN    | percentile     | not applicable | 0.9       |
| KNet     | percentile     | not applicable | 0.9       |
| SwinIR   | percentile     | not applicable | 0.9       |

and in the configuration reported here the allocation is formed for a single multiplicative gain proposal, so that truth value is the constant zero and the term  $-C_5 w_5 r_5$

TABLE S5. Feature tensor read by the cooperation head for each backbone. All heads share the same two layer design and differ only in the number of input channels.

| Backbone | Feature tensor              | Channels |
|----------|-----------------------------|----------|
| NAFNet   | First encoder stage         | 40       |
| KNet     | First encoder stage         | 32       |
| DnCNN    | First convolution           | 228      |
| SwinIR   | Shallow feature convolution | 138      |

vanishes whatever  $w_5$  is. The reported implementation therefore has four active weighted rules and a trainable base, with the fifth rule configured and inactive and the six thresholds fixed, which is the inventory Section V-I of the main paper measures.

The repair also tightens the theory. Because the allocation is affine in the rule truth values with signed coefficients and no clipping intervenes, the Lipschitz constant of Theorem 1 of the main paper is attained rather than merely an upper bound.

This is also what the trained model shows and not only a property of the formula. Measured on the released weights, the allocation now spans 0.556 to 0.793 with mean 0.726 and standard deviation 0.050, and 0.00 percent of pixels sit at either end of the clamp, against the whole image before the repair, direct confirmation that no pixel saturates in practice.

### C. Reproducing the earlier behavior

Setting the environment variable LEGACY\_SATURATING\_ALLOCATION to 1 restores the original computation exactly. We verified

this against a copy of the pre revision source on twenty random inputs, comparing the allocation maps for equality and the rule traces for agreement. Both matched. The earlier numbers can therefore be recovered from the released code, which is why the switch exists.

#### S5. Leave one out procedure for the clinical properties

The results section of the main paper reports the contribution of each clinical property. The procedure is a leave one out ablation. At test time, one property at a time is switched off by setting its score to the satisfied value and its failure map to zero, and the whole test set is re evaluated with that property held-out. This is the same kind of one at a time removal used for the rule constants and for the components of the method, applied here to the six clinical properties instead.

#### S6. Full per cell spreads

The results table of the main paper prints each clinical cell as a mean, with the robustness test folded into the choice of bold, and drops the per seed spread to keep the table one page. Table S6 here restores that spread for every cell, four backbones by three blocks by six measures, each as mean plus or minus one standard deviation across three seeds, generated directly from the same result files.

#### S7. The cooperation map, in full

The main paper reports that the map produced by the head reading the backbone features does not predict the error of that backbone. This section gives the rest of the analysis.

Two measures were computed over the 173 test images. The quantity correlated with the absolute error of the backbone output is the demand map  $\mathbf{u} = \mathbf{1} - \mathbf{c}$ , which is high where the wrapper is told to act. A demand map that tracked backbone error would give a positive correlation. The measured rank correlation is -0.258, so the ordering runs the other way. The sparsification error, the area between the curve obtained by removing the pixels the map calls least reliable and the curve obtained by removing the pixels that are actually worst, is 0.776. A perfect ordering scores zero on the second measure and one on the first.

The reason is structural rather than a matter of tuning. The head is trained only through the cooperation term of the objective, which rewards agreement between the map and the estimated correction potential. That potential is itself a learned quantity. A map supervised only against another learned map has no path by which the error of the backbone could reach it, so there is no reason for it to acquire the meaning the original name implied.

Repairing this would require a term that regresses the map on the measured backbone error. Such a term

would compete with the cooperation term rather than complement it, because one pulls the map toward the error of the backbone and the other pulls it toward where the correctors want to act. The relative weight of the two would have to be selected, and every result in this paper would have to be recomputed. We therefore report the measurement and leave the repair as the clearest piece of future work this revision has identified.

#### S8. The adapted transfer protocol of the original submission, a test selected diagnostic

These numbers are not an estimate of generalization. Each fold keeps the epoch that scores best on the subject it then reports, so they are selected on the data they describe. They are here as a record of what the original submission did and of the gap between the two protocols, and the main paper reports the frozen protocol throughout.

The cross dataset numbers of the original submission were not produced by the no adaptation protocol the text described. The script that produced them resumed the corrector trained on PKU37 and trained it further on the target dataset in a leave-one-subject-out design. We rerun that protocol here with the selected NAFNet checkpoint so that the gap between the two protocols can be read directly, and we describe it exactly.

#### A. Protocol

Each fold holds out one subject, seventeen of Duke2013 or fifteen of Duke17 subjects form the training set, and the held-out subject serves as both the validation subject and the reported one. Training resumes from the PKU37 checkpoint of the tissue-bounded gate setting at seed 0 and runs for fifteen epochs with batch size two. The backbone weights and its normalization statistics are frozen. The wrapper and the cooperation head that reads the backbone features are trainable. The learning rates are  $2 \times 10^{-5}$ ,  $1 \times 10^{-4}$  and  $6 \times 10^{-5}$  for the corrector, potential and negotiator, five to ten times below the PKU37 schedule of Table S2, the fidelity term enters at epoch five, and an L2 penalty of weight 0.1 holds every trainable weight near its PKU37 value, which is elastic weight consolidation without the Fisher weighting. A background correction variance penalty of weight 10 is also active. The checkpoint kept for each fold is the epoch with the highest validation score, and the validation image of a fold is its held-out subject. The adapted numbers are therefore selected on the subject they are evaluated on. That makes them optimistic rather than an upper bound in any proved sense, and they should be read as a diagnostic of what the original submission did. The same was true of the original submission. We did not repair this by reselecting, since no unbiased checkpoint rule can be recovered from folds that were already trained this way.

TABLE S6. Every clinical cell of Tables 3 and 4 of the main paper, with its full mean plus or minus one seed standard deviation restored. Each cell here is the one those tables show as a mean alone, and bold marks a mean that exceeds its spread.

| Backbone                   | $\Delta$ CNR      | $\Delta$ TCI      | $\Delta$ EPI     | $\Delta$ BS      | $\Delta$ ENL      | $\Delta$ SNR      |
|----------------------------|-------------------|-------------------|------------------|------------------|-------------------|-------------------|
| PKU37 test set, 173 images |                   |                   |                  |                  |                   |                   |
| NAFNet                     | +3.49 $\pm$ 0.19  | +1.91 $\pm$ 0.51  | +1.20 $\pm$ 0.02 | +3.89 $\pm$ 0.45 | +23.74 $\pm$ 0.38 | +11.10 $\pm$ 0.31 |
| DnCNN                      | +11.30 $\pm$ 0.52 | +0.99 $\pm$ 1.62  | +4.14 $\pm$ 0.16 | -1.44 $\pm$ 0.87 | +57.14 $\pm$ 7.63 | +22.95 $\pm$ 2.28 |
| SwinIR                     | +7.33 $\pm$ 0.81  | +11.45 $\pm$ 0.86 | +2.68 $\pm$ 0.16 | +1.82 $\pm$ 0.37 | +28.32 $\pm$ 1.30 | +11.54 $\pm$ 0.26 |
| KBNet                      | +6.93 $\pm$ 0.98  | +5.92 $\pm$ 1.81  | +0.00 $\pm$ 0.07 | +2.58 $\pm$ 1.35 | +25.30 $\pm$ 2.17 | +7.95 $\pm$ 0.36  |
| Duke17, no adaptation      |                   |                   |                  |                  |                   |                   |
| NAFNet                     | +7.51 $\pm$ 0.21  | +4.00 $\pm$ 1.08  | +1.37 $\pm$ 0.07 | +4.15 $\pm$ 0.16 | +21.35 $\pm$ 1.78 | +5.43 $\pm$ 0.80  |
| DnCNN                      | +12.19 $\pm$ 0.72 | -1.47 $\pm$ 3.15  | +3.49 $\pm$ 0.15 | -2.24 $\pm$ 1.55 | +39.42 $\pm$ 5.44 | +13.55 $\pm$ 1.35 |
| SwinIR                     | +8.68 $\pm$ 1.12  | +13.26 $\pm$ 0.91 | +2.69 $\pm$ 0.24 | +1.90 $\pm$ 1.00 | +29.46 $\pm$ 0.66 | +11.60 $\pm$ 0.29 |
| KBNet                      | +5.34 $\pm$ 0.87  | +7.19 $\pm$ 1.57  | -0.50 $\pm$ 0.19 | +1.61 $\pm$ 0.09 | +7.49 $\pm$ 3.76  | -1.81 $\pm$ 1.80  |
| Duke2013, no adaptation    |                   |                   |                  |                  |                   |                   |
| NAFNet                     | +7.37 $\pm$ 0.06  | +1.41 $\pm$ 0.87  | +0.85 $\pm$ 0.04 | +0.51 $\pm$ 0.16 | +28.80 $\pm$ 2.25 | +9.29 $\pm$ 1.25  |
| DnCNN                      | +12.73 $\pm$ 1.41 | -2.21 $\pm$ 2.71  | +2.99 $\pm$ 0.25 | -5.85 $\pm$ 0.87 | +49.91 $\pm$ 7.29 | +17.59 $\pm$ 2.04 |
| SwinIR                     | +9.26 $\pm$ 1.20  | +11.86 $\pm$ 1.10 | +2.55 $\pm$ 0.20 | +2.30 $\pm$ 1.69 | +30.90 $\pm$ 1.02 | +13.31 $\pm$ 0.20 |
| KBNet                      | +7.16 $\pm$ 0.94  | +4.83 $\pm$ 1.32  | -0.35 $\pm$ 0.09 | +3.06 $\pm$ 1.03 | +8.70 $\pm$ 4.55  | -0.90 $\pm$ 2.14  |

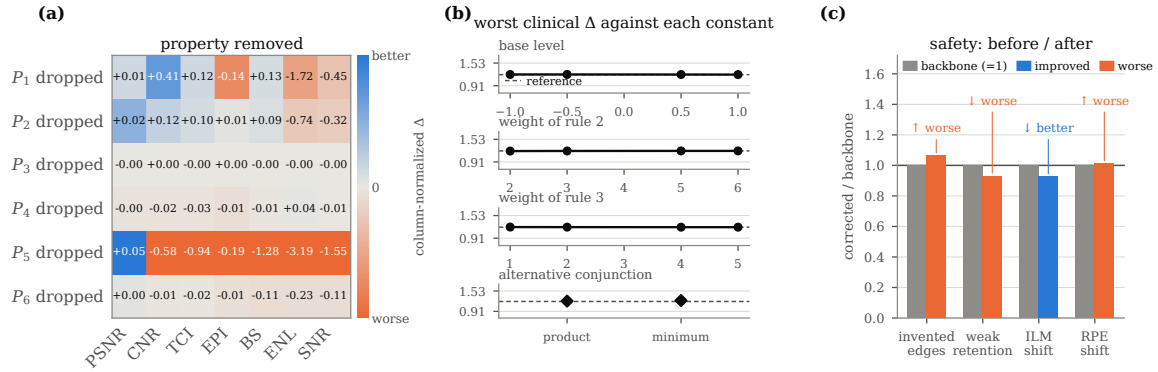


FIGURE S1. The three studies of the main paper drawn together. Left, removing one clinical property at a time, in points of the measure. Only  $P_5$ ,  $P_1$  and  $P_2$  move any measure by more than a quarter point, so  $P_3$ ,  $P_4$  and  $P_6$  are close to inert here. Middle, the fourteen variations of the rule constants, none of which turns a clinical measure negative, with the base allocation level the only visible effect. Right, the safety analysis, each of the invented edge rate, weak structure retention and the two boundary shifts shown as the ratio of corrected to backbone value, one meaning unchanged, the arrow giving the direction and the word whether it is better or worse for that measure. Three ratios sit on the worse side, though the retinal pigment epithelium shift moves one percent and is not significant. All panels are the selected NAFNet checkpoint on the same 173 test images.

## B. Result

Table S7 gives both target sets. The no adaptation column is the mean over the three seeds reported in the main paper, while the adapted column is one seed per fold, so the two are not seed matched and the comparison is indicative.

Two patterns appear on both scanners, and neither can be read as an effect of adaptation for the reason given above. Contrast is higher, by 3.2 points on Duke17 and 8.1 on Duke2013, on every subject. Fidelity and signal to noise ratio and signal to noise ratio are lower than the frozen model by a further 0.06 and 0.10 dB, with one and two subjects losing more than 1 dB, and a signal to noise ratio 2.5 and 14.6 points lower. The mean absolute

correction rises from 0.0104 to 0.0121 on Duke17 and from 0.0118 to 0.0164 on Duke2013. The other measures do not agree between the two sets. On Duke17 the tissue contrast index reads +2.7 against +4.0 percent while the speckle measure is higher, and no measure is below zero. On Duke2013 the tissue contrast index, edge preservation and boundary sharpness are all higher, the speckle measure moves one way on eight subjects and the other on ten, and the signal to noise ratio is 5.3 percentage points lower on fifteen of eighteen subjects. We do not attach significance language to any of these, for the reason below.

We draw no conclusion about what adaptation achieves. These numbers are kept only to document the

TABLE S7. NAFNet, adapted protocol over one leave-one-subject-out fold per subject, against the no adaptation protocol of Table 4 of the main paper, whose column is the mean over three seeds, so the two columns are not seed matched. Clinical rows are percentage change from backbone output to corrected output. The spread is the standard deviation across subjects, and the count is the number of subjects on which the measure improved. No  $p$  value is given, because every checkpoint was chosen on the subject it is then scored on, so a test on those paired values would not carry its usual meaning.

| Measure               | No adaptation | Adapted            | Improved |
|-----------------------|---------------|--------------------|----------|
| Duke17, 16 subjects   |               |                    |          |
| $\Delta$ PSNR (dB)    | -0.17         | $-0.23 \pm 0.44$   | 6 of 16  |
| $\Delta$ CNR          | +7.51         | $+10.68 \pm 1.55$  | 16 of 16 |
| $\Delta$ TCI          | +4.00         | $+2.69 \pm 4.16$   | 12 of 16 |
| $\Delta$ EPI          | +1.37         | $+1.49 \pm 0.78$   | 15 of 16 |
| $\Delta$ BS           | +4.15         | $+4.13 \pm 5.18$   | 13 of 16 |
| $\Delta$ ENL          | +21.35        | $+24.29 \pm 17.72$ | 15 of 16 |
| $\Delta$ SNR          | +5.43         | $+2.95 \pm 8.97$   | 9 of 16  |
| Duke2013, 18 subjects |               |                    |          |
| $\Delta$ PSNR (dB)    | -0.35         | $-0.45 \pm 0.47$   | 4 of 18  |
| $\Delta$ CNR          | +7.37         | $+15.43 \pm 4.09$  | 18 of 18 |
| $\Delta$ TCI          | +1.41         | $+5.15 \pm 3.75$   | 17 of 18 |
| $\Delta$ EPI          | +0.85         | $+0.90 \pm 0.89$   | 15 of 18 |
| $\Delta$ BS           | +0.51         | $+2.96 \pm 2.97$   | 14 of 18 |
| $\Delta$ ENL          | +28.80        | $+9.23 \pm 28.61$  | 8 of 18  |
| $\Delta$ SNR          | +9.29         | $-5.29 \pm 13.97$  | 3 of 18  |

procedure the original submission used. Every checkpoint was chosen on the subject it is then scored on, and the comparison is not seed matched, so neither the differences above nor the  $p$  values are unbiased evidence that adaptation helps or hurts any measure. They cannot support a confirmatory claim in either direction. The main conclusions of this paper rest on the frozen transfer of Table 4 of the main paper, which gives six positive measures on both sets with nothing fitted on the target data.